NIRIKSHAN

Product Requirements Document

AI-assisted building deterioration monitoring and inspection prioritisation for Mumbai

Prepared for SEWA FIRST Innovation Challenge 2026

Recommended category: REG-009-OP - Safety & Disaster Management

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Core principle: NIRIKSHAN does not decide whether a building is safe or unsafe. It detects visible change over time and helps engineers decide what to inspect first.

1. Summary

NIRIKSHAN turns repeat smartphone photographs of ageing buildings into a longitudinal record of visible deterioration. It aligns comparable images of the same building element, measures visible change, and ranks cases by how urgently they should be reviewed by a structural engineer. It does not assign a structural safety rating, predict collapse, or replace formal inspection.

One-line pitch: NIRIKSHAN uses repeat smartphone photographs to detect how visible damage in ageing Mumbai buildings changes over time, helping engineers identify which buildings should be revisited first between formal inspections.

2. Problem

Mumbai's Building Repairs and Reconstruction Board (MBRRB, under MHADA) conducts a pre-monsoon survey of old cessed buildings. Published 2026 reports state that 82 highly dangerous buildings in South Mumbai housed 2,736 tenants, and 43 of those buildings had also appeared on the previous year's list. A 2020 report put the city's cessed-building stock above 14,000 and noted that the board operates an emergency control room for distress calls.

These figures are contextual evidence only and must be re-verified against MHADA/MBRRB primary notices before final submission.

The operational gap:

- Formal structural inspections are periodic, while visible condition can change between inspection cycles, especially across monsoon periods.
- Engineering teams cannot revisit thousands of buildings at high frequency.
- Residents and field staff may notice visible changes, but observations are rarely captured in a consistent, comparable form.
- A single photo says little about progression. The more useful question is what has changed since the last comparable observation.
- There is no low-cost, standardised longitudinal visual record that can automatically surface the most changed cases for human review.

3. Goals and non-goals

Goals

1. Detect visible distress from smartphone images, beginning with cracks in the MVP and adding spalling as a secondary feature if time allows.
2. Register repeat images of the same predefined building element and separate real visible deterioration from lighting, camera, weather and viewpoint differences.
3. Measure relative change in visible damage and produce an explainable Inspection Priority Score for engineer review.
4. Maintain an auditable per-building timeline of source images, model outputs, score history and engineer decisions.
5. Work with commodity smartphones and open-source software, with no specialised capture hardware required for the MVP.
6. Prefer uncertainty escalation and human review over confident automated conclusions when image quality or alignment is poor.

Non-goals

- Predicting collapse or assigning a building-level safety rating.
- Replacing structural audits, engineering judgement or statutory inspection procedures.
- Inferring hidden corrosion, foundation condition or internal structural defects that are not visible in photographs.
- Reporting precise physical crack width, length or area from an uncalibrated phone image.
- Drone capture or 3D reconstruction in the MVP; these remain roadmap capabilities.

4. Users

Persona	Need	Primary use
Inspection engineer (MBRRB, MHADA, BMC or partner body)	Know which cases to inspect first and see evidence of change	Ranked queue, before/after review, building timeline
Field capturer (junior engineer, trained volunteer, NSS/NCC student)	Capture repeatable, usable exterior photographs quickly	Guided capture app with previous-photo overlay and quality gates
Resident or society member (later phase)	Submit a visible change for consideration without creating a public safety claim	Simple report flow routed to human review
Administrator	Manage buildings, users, thresholds and audit history	Admin console

5. Scope

MVP - core end-to-end loop (Build stage: 12 Oct to 10 Nov 2026)

- Building registration and baseline photo set.
- Predefined capture points for approximately planar building elements such as wall faces, slab undersides or column faces.
- Guided capture using the previous image as a semi-transparent overlay.
- Image-quality and viewpoint/parallax checks with recapture prompts.
- Crack segmentation as the primary P0 visual-damage model.
- Image registration between comparable visits using feature matching and homography.
- Relative crack-change metrics between aligned captures.
- Transparent heuristic Inspection Priority Score and ranked engineer review queue.
- Engineer dashboard with before/after visual evidence and building timeline.
- Staged and archival validation with at least one practising structural/civil engineer reviewing sample outputs.

MVP if time allows (P1)

- Spalling segmentation and change metrics.
- Map view and PDF export.
- Offline capture with later sync and on-device privacy blurring.
- Verified building metadata such as age, repair history and prior inspection category.

Phase 2

- Exposed reinforcement detection.
- Moisture staining and efflorescence detection.
- Balcony, slab-edge and chajja-specific analysis.
- Monsoon Watch recheck list based on recent visual change and heavy-rain periods.
- Citizen reporting into a human-review queue.

Phase 3

- Structure-from-motion facade reconstruction.
- Ingestion of imagery from authorised inspection drones.
- Integration with MBRRB/MHADA/BMC workflows and control-room systems.

6. Core user flows

6.1 Register a building

7. Engineer or administrator creates a building record: ID, address, ward, age, previous inspection category if known, and notes.
8. Field capturer records a baseline set of exterior images at predefined capture points.
9. Each capture point is tied to an approximately planar physical element and a recommended viewpoint.
10. The app checks blur, exposure, subject coverage and basic viewpoint quality before accepting the photo.

6.2 Periodic capture

11. Capturer selects the same building and capture point.
12. The previous accepted photo appears as a semi-transparent overlay.
13. The app shows live alignment guidance and blocks capture when blur, exposure, distance or viewpoint/parallax is beyond threshold.
14. If the capture is acceptable, the image is stored locally and uploaded when connectivity is available.

6.3 Analysis

15. Register the new image to the previous image using feature matching and homography.
16. Normalise illumination and verify registration confidence.
17. Run crack segmentation on both images; run spalling segmentation only when the P1 module is enabled.
18. Compare aligned damage masks and compute relative change metrics.
19. Reject unreliable comparisons rather than forcing a score.
20. Recalculate the building Inspection Priority Score and store the evidence trail.

6.4 Engineer review

21. Engineer opens the ranked review queue.
22. Engineer sees the before/after images, aligned overlays, highlighted changes and score contributors.
23. Engineer marks each finding as confirmed, false alarm, uncertain or site visit required.
24. Feedback is retained for evaluation, threshold calibration and later model improvement.

7. Functional requirements

Priority key: P0 = MVP must-have; P1 = build if the P0 loop is working end to end; P2 = later phase.

7.1 Capture app

ID	Requirement	Priority
CAP-1	Capture images with building ID, capture point and timestamp; location may be recorded only where permitted.	P0
CAP-2	Show the previous accepted image as an adjustable overlay during capture.	P0
CAP-3	Real-time quality checks for blur, exposure, subject coverage and approximate distance.	P0
CAP-4	Reject large viewpoint/parallax differences that make planar registration unreliable.	P0
CAP-5	Allow a reference scale marker in	P1

	frame to support approximate physical measurements.	
CAP-6	Offline capture with later sync.	P1
CAP-7	Blur faces and vehicle number plates on device before upload.	P1

7.2 Damage detection

ID	Requirement	Priority
DET-1	Segment crack pixels and output relative image-space features such as mask area, skeleton length, orientation and branching.	P0
DET-2	Report change in crack extent as relative percentages between aligned captures.	P0
DET-3	Segment spalling regions and report relative area change.	P1
DET-4	Segment exposed reinforcement.	P2
DET-5	Segment moisture staining and efflorescence.	P2
DET-6	Report a confidence value and uncertainty flag for each detection.	P0

Measurement rule: Without a scale marker or calibrated geometry, the system must not present phone-derived crack width, length or area as precise physical measurements. It should report relative image-space change. With a reference scale, approximate physical measurements may be shown and clearly labelled as approximate.

7.3 Image registration and change detection

ID	Requirement	Priority
CHG-1	Register new and previous images of the same predefined capture point using feature matching and homography.	P0
CHG-2	Normalise illumination before	P0

	comparison.	
CHG-3	Output new, extended, reduced and unchanged visible damage regions.	P0
CHG-4	Reject comparisons below alignment-confidence threshold and request recapture.	P0
CHG-5	Reject capture pairs with excessive viewpoint or parallax difference.	P0
CHG-6	Suppress apparent changes explained by shadows, rain streaks, vegetation and movable clutter.	P1
CHG-7	Evaluate a learned Siamese/change-detection model only if classical registration plus mask comparison is inadequate.	P2

Scope note: Homography is appropriate for approximately planar elements captured from similar viewpoints. The MVP must not claim reliable whole-building alignment across large viewpoint changes or strong depth variation.

7.4 Inspection Priority Score

ID	Requirement	Priority
SCR-1	Compute a 0-100 review-order score from visible change features using configurable weights.	P0
SCR-2	Initial pilot bands: 0-30 Routine, 31-60 Review recommended, 61-80 High, 81-100 Urgent human review.	P0
SCR-3	Show the top contributing factors and linked image evidence for every score.	P0
SCR-4	Use verified metadata such as building age, previous inspection category and repair history only as P1 contextual inputs.	P1
SCR-5	Include rate-of-change only when at least three comparable captures are	P1

	available.	
SCR-6	Calibrate score thresholds and weights using validation data and engineer feedback.	P0

Interpretation: This is a transparent heuristic inspection-priority model, not a structural-risk equation. The weights are not safety coefficients. They determine review ordering only. The product must never use the terms collapse probability, safe or unsafe. Initial score bands are pilot thresholds and must be calibrated during validation.

7.5 Dashboard

ID	Requirement	Priority
DSH-1	Ranked queue of buildings with score, band and reason summary.	P0
DSH-2	Building page with chronological timeline of images, scores and engineer decisions.	P0
DSH-3	Side-by-side and overlay before/after views with detected changes highlighted.	P0
DSH-4	Map view coloured by review band.	P1
DSH-5	Engineer feedback controls: confirmed, false alarm, uncertain, needs visit.	P0
DSH-6	Export a building review report as PDF.	P1

7.6 Citizen reporting and Monsoon Watch (Phase 2)

ID	Requirement	Priority
CIT-1	Residents may submit exterior photos with a short description.	P2
CIT-2	Submissions are matched to a building and routed to human review; they never directly alter the score.	P2

MON-1	Before and after heavy-rain periods, generate a recheck list using recent visible-change history and rainfall context.	P2

8. AI and ML requirements

- Primary MVP model: crack segmentation. Candidate architectures include U-Net, DeepLabV3+, SegFormer or YOLO segmentation; final choice must be based on validation rather than model popularity.
- Secondary P1 model: spalling segmentation, added only after the crack-change loop works end to end.
- Registration: classical feature matching and homography first; learned registration/change methods are a later option if classical methods fail on representative data.
- Training data: combine suitable public structural-damage datasets with locally collected and labelled photographs. Target 500 to 1,500 carefully annotated local images where feasible.
- Annotation: segmentation masks for visible damage, plus capture-pair labels for changed/unchanged regions.
- Augmentation: lighting, blur, perspective, colour and sensor-noise variations to approximate different phones and field conditions.
- Explainability: every flagged change must be visually localised on the source image and linked to the before/after evidence.
- Human in the loop: engineer feedback is logged and used to measure false alarms, calibrate thresholds and guide later retraining.

9. Non-functional requirements

- Accuracy over automation: uncertain cases are sent to human review rather than silently assigned a confident priority.
- Performance: analyse one accepted image pair in under 60 seconds on a single GPU or reasonable CPU setup.
- Usability: a trained capturer should complete one capture point in under two minutes.
- Offline tolerance: field capture must work without connectivity and sync later.
- Privacy: store exterior imagery only; blur faces and number plates where feasible; restrict access by role.
- Auditability: keep an immutable history of images, model versions, scores, threshold versions and engineer decisions.
- Cost: no specialised hardware is required for the MVP; commodity smartphones and open-source tools are sufficient.
- Safety language: no automated output may be presented as a structural safety verdict.

10. Success metrics

The values below are targets to be measured, not achievements. They must not be quoted as results until validation is complete.

Metric	Target	How measured
Crack segmentation IoU	0.75 or higher	Against manual masks on a held-out

		test set
Spalling segmentation F1 (P1 only)	0.80 or higher	Against manual labels on a held-out set
Registration success rate	90% or higher of capture pairs	Fraction of pairs passing alignment and parallax-quality gates
Change detection recall on staged worsening	85% or higher	Staged and archival before/after test set
False alarm rate on unchanged pairs	10% or lower	Same test set, on known unchanged pairs
Ranking Recall@20%	At least 85% of deliberately worsened cases appear in the top 20% of the review queue	100-image-pair ranking test
Engineer usefulness agreement	70% or more of flagged changes rated useful for deciding whether further inspection is warranted	Blind or semi-blind review by at least one practising structural/civil engineer

11. Validation plan

Real structural deterioration is slow, so validation must not depend on waiting for natural damage during the competition period.

25. Controlled surfaces: safely create or simulate progressive visible marks/damage on panels or non-critical surfaces and photograph them repeatedly under varied phones, lighting and viewpoints.
26. Archival pairs: obtain before/after images of the same building element from renovation records, society archives or other lawful sources where the nature of visible change is known.
27. Accessible buildings: with permission, repeatedly photograph exterior regions of five to ten older buildings. Do not describe any participating building as unsafe.
28. Manual annotation: two annotators independently label a subset of the test data; report agreement where feasible.
29. Ranking test: mix unchanged and deliberately worsened pairs, then measure Recall@20% and false alarms.
30. Engineer review: involve at least one practising structural or civil engineer before final validation. The engineer reviews damage categories, score logic and a sample of system outputs, and provides written feedback.
31. Threshold calibration: treat score bands and heuristic weights as pilot values, then adjust them using validation evidence and engineer feedback without presenting them as safety thresholds.

12. Timeline against SFIC 2026

SFIC stage	Dates	NIRIKSHAN deliverables

Ideate	to 11 Oct 2026	Idea submission, revised PRD, problem statement, validation plan, identify structural/civil engineer adviser
Build and Evaluate	12 Oct to 10 Nov	Guided capture prototype, crack segmentation, image registration, crack-change metric, priority queue, dashboard, first staged tests; add spalling only if core loop is stable
Validate and Test	11 Nov to 19 Nov	Field captures on accessible buildings, ranking test, threshold calibration, engineer review and written feedback
Final Select	20 Nov to 26 Nov	Final report, live end-to-end demo, measured results, limitations and honest error analysis

Suggested internal sprints during Build:

- Week 1: data collection, annotation setup, baseline crack model, guided capture overlay.
- Week 2: image registration, parallax/quality rejection, first before/after crack-change output.
- Week 3: priority score, dashboard, building timeline and engineer-review controls.
- Week 4: staged tests, threshold calibration, error analysis and demo hardening. Add spalling only if the P0 loop is complete.

13. Risks and mitigations

Risk	Impact	Mitigation
Insufficient labelled damage data	Weak model quality	Prioritise cracks; use public data plus local annotation. Keep spalling at P1 until data quality is adequate.
Lighting and camera differences look like change	False alarms	Illumination normalisation, alignment-confidence gate, controlled tests across phones and conditions.
Large viewpoint/parallax difference	Misregistration and false change	Guided overlay, predefined planar capture points, viewpoint rejection and recapture prompts.
Poor capture consistency	Unusable comparisons	Quality gates, overlay guidance and explicit capture-point instructions.
Visible condition does not reveal	False reassurance	State limitation prominently; never

hidden structural problems		output a safety verdict; require human engineering review.
False low priority on a deteriorating case	Safety and reputational risk	Conservative uncertainty handling, human review, periodic audit of low-score cases and engineer calibration.
Building access or ownership disputes	Cannot collect field data	Use legally accessible exterior views and obtain owner/society/institutional permission where required.
Insufficient time to observe natural deterioration	Weak validation story	Use staged and archival pairs, clearly labelled as staged/archival rather than real-time deterioration.
Scope creep	Unfinished prototype	Freeze P0 until the full capture-to-review loop works reliably.

14. Ethics, privacy and safety

- The product supports engineers and does not replace them.
- Outputs use inspection-priority and review language, never structural-safety verdicts.
- Residents are not penalised, publicly named or alarmed based solely on automated output.
- Personal data visible in exterior imagery should be minimised or blurred; interior imagery is out of scope.
- Building scores and raw review findings are restricted to authorised users to reduce panic and property-value harm from unverified automated results.
- Any public demonstration should use consented, anonymised or staged examples unless official permission has been obtained.

15. Open questions

32. Will MBRRB, MHADA, BMC or another civic body review sample outputs or provide anonymised historical material?
33. Which practising structural/civil engineer can advise on visible damage categories and review final outputs?
34. Which accessible buildings or controlled surfaces can the team photograph repeatedly, and with whose permission?
35. What is the final SFIC judging rubric, and how strongly does it weight a validated working prototype?
36. What compute is available for training and evaluation (local GPU, Colab or institutional cluster)?
37. Can a consistent small reference marker be included in controlled capture points for approximate scale where permitted?

Submission category recommendation: REG-009-OP - Safety & Disaster Management. The project can also be described as an urban resilience application, but its primary value proposition is inspection prioritisation for disaster-risk reduction.

16. Appendix: Inspection Priority Score definition (draft)

$$P = w1*C + w2*S + w3*R + w4*W + w5*T$$

- C: normalised relative change in crack extent between accepted aligned captures.
- S: normalised relative change in spalling area (P1).
- R: exposed reinforcement change indicator (Phase 2).
- W: moisture-deterioration change indicator (Phase 2).
- T: estimated rate of visible change across at least three comparable captures.

Initial weights are hand-set only to order the review queue and must be calibrated with engineer feedback and validation data. They are not structural-safety coefficients. Missing terms are excluded and remaining weights are renormalised. With only two comparable captures, T is excluded. Every score must show its contributing factors and link directly to the images behind them.

17. MVP acceptance test

The MVP is complete only when the following loop works end to end on unseen test pairs:

CAPTURE -> QUALITY GATE -> ALIGN -> SEGMENT CRACKS -> MEASURE RELATIVE CHANGE -> PRIORITISE -> ENGINEER REVIEW

Spalling, mapping, PDF export, citizen reporting, drone imagery and 3D reconstruction do not block MVP completion. A smaller complete system is preferred to a larger partially working one.